One finite nonzero boundary derivative forces bi-Lipschitz conjugacy (Proposition 7.2)

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Editorial note. One Proposition 7.2 counted; author text and all current core retained unchanged.

Editorial note. Original short-remainder estimate retained exactly, without mathematical certification.

Editorial note. r002 releases the article-class legacy cal alias before the unchanged native author macro definition; mathematical text untouched.

Editorial note. r003 restores published equation numbers 7.1–7.5 and suppresses forced publisher page breaks in the independent wrapper. Mathematical text unchanged.

Editorial scope. One whole Proposition 7.2; native mathematics unchanged; source estimate retained with notice; standard wrapper and five faithfully reconstructed cited references.

Allowed standard prerequisites (editorial).

- Classical hyperbolic group boundary and cone types. Cannon [Can84]; Bridson–Haefliger [BH99], p.455; original lines 1391–1410. Boundary/geodesic conventions retained in full.
- Classical conjugacy extension for convex co-compact Kleinian representations. Marden [Mar74], Anderson [And98], p.32; original lines 2620–2622. Provides the ambient Hölder conjugacy stated by the authors.
- Chain rule, geometric series and Arzelà–Ascoli. Standard analysis; full current bounded-distortion and rescaling core retained. Elementary analysis and compactness tools only, not replacement for author core.

Source reference. Original section 3: local-coloring-linear-logic/AHL265/sources/author-published.tex, source lines 1216–1225. Backward naming pointer only: the selected sentence itself defines the distinct-boundary-pair notation.

Source, scope and retained wording

Maria Beatrice Pozzetti and Andrés Sambarino, *Metric properties of boundary maps, Hilbert entropy and non-differentiability*, Annales Henri Lebesgue 9 (2026), 205–260. Exact final published native text, `https://doi.org/10.5802/ahl.265`; CC BY 4.0, `https://creativecommons.org/licenses/by/4.0/`. Original printed pages 248–252 contain the selected result and its full proof.

Only whole Proposition 7.2 is counted. Both circle and convex co-compact Kleinian settings, the Lipschitz-compatible-cover hypotheses, complete supporting Lemma 7.3 and complete main proof are retained. The derivative must be finite and nonzero; it is a complex derivative in the Kleinian case. Lemma 7.4, which establishes covers for further applications, is not assumed as a new theorem here: existence of these covers is an explicit hypothesis of Proposition 7.2.

Author wording notice. The final estimate in Lemma 7.3, original native lines 2716–2721, retains the author's factor involving $e^{-mL\lambda}$ and the cited preceding estimate exactly. No alternative estimate, mathematical repair or certification is supplied. The complete written proof is present; this notice distinguishes faithful source transcription from an independent correctness judgment.

Difficulty and identity boundary. The retained H2 core combines distortion along group words, paired rescalings at one differentiability point, bi-Lipschitz compactness limits and equivariant globalization. Arzelà–Ascoli and classical hyperbolic boundary conventions are standard inputs. Saved in-flight identities end at 14 October 2026, 17:36:45 UTC; newer overlap remains unknown. Preparation is not final global-unique admission.

4.1 Gromov hyperbolic groups and cone types

Let $\Gamma = \langle S \rangle$ be a finitely generated non-elementary Gromov hyperbolic group, and recall from Section 3 that we denote by $\partial^2\Gamma$ the set of distinct pairs in its Gromov boundary $\partial\Gamma$.

Definition 4.1. A divergent sequence $\{\gamma_n\}_{n\in\mathbb{N}} \subset \Gamma$ converges to a point $x \in \partial\Gamma$ *conically* if for every $y \in \partial\Gamma - \{x\}$ the sequence $(\gamma_n^{-1}y, \gamma_n^{-1}x)$ remains on a compact set of $\partial^2\Gamma$.

Remark 4.2. It is easy to verify that a sequence $\{\gamma_n\}_{n\in\mathbb{N}}$ converges conically to $x \in \partial\Gamma$ if and only if it lies in an uniform neighborhood of any geodesic ray $(\alpha_n)_0^\infty$ converging to x, namely there exists $K > 0$ and a subsequence $\{\alpha_{n_k}\}$ such that for all k one has $d_\Gamma(\alpha_{n_k}, \gamma_k) < K$.

Given $\gamma \in \Gamma$ we denote by $\mathcal{C}(\gamma)$ the *cone type* of $\gamma \in \Gamma$, namely

$$\mathcal{C}(\gamma) := \{h \in \Gamma \,|\, d(e, \gamma h) = d(e, \gamma) + d(e, h)\}\,.$$

Cannon [Can84] showed the set of cone types of a Gromov hyperbolic group is finite, see for example Bridson–Haefliger's book [BH99, p. 455]. We denote by $\mathcal{C}_\infty(\gamma) \subset \partial\Gamma$ the set of points x that can be represented by a geodesic ray contained in $\mathcal{C}(\gamma)$.

Either Γ has boundary homeomorphic to a circle, or it is a Kleinian group. In the first case we let

$$\rho,\bar\rho : \Gamma \to \mathrm{Diff}^{1+\nu}(\mathbb{S}^1)$$

be Hölder conjugated to action of Γ on its boundary; if instead $\Gamma < \mathsf{PSL}(2,\mathbb{C})$ is a Kleinian group we let $\rho,\bar\rho : \Gamma \to \mathsf{PSL}(2,\mathbb{C})$ be two convex co-compact representations that lie in the same connected component of the subset of the character variety $\mathfrak{X}(\Gamma, \mathsf{PSL}(2,\mathbb{C}))$ consisting of convex co-compact representations.

We let X be either the circle or $\partial\mathbb{H}^3$. To simplify notation we will denote the action of $\gamma \in \Gamma$ on X via ρ by γ, the action via $\bar\rho$ by $\bar\gamma$, and the limit sets of ρ and $\bar\rho$ by $\partial\Gamma, \bar\partial\Gamma \subset X$ respectively.

In both situations there exists a Hölder-continuous map

$$\Xi : X \to X$$

conjugating ρ and $\bar\rho$. Indeed while in the surface case this holds by definition, in the Kleinian case this is a theorem by Marden [Mar74], see also Anderson's survey [And98, p. 32]: the equivariant limit map $\Xi : \partial\Gamma \to \partial\Gamma$ conjugating the actions ρ and $\bar\rho$ on their respective limit sets extends to a Γ-equivariant, Hölder continuous homeomorphism of the whole Riemann sphere $\partial\mathbb{H}^3$. We study differentiability points of Ξ with oblique derivative.

We let d be either a visual distance on X (in the complex case) or a distance inducing the chosen C^1 structure on the circle $\mathbb{S}^1$.

Definition 7.1. An action ρ admits a *Lipschitz-compatible cover* if there exists a finite open cover $\mathcal{B}$ of X and a map $\Gamma \to \mathcal{B}$, $\gamma \mapsto \mathcal{B}_\infty(\gamma)$ such that:

(i) for any $a, b \in \Gamma$ so that $|ab| = |a| + |b|$ one has

  (a) $b\mathcal{B}_\infty(ab) \subset \mathcal{B}_\infty(a)$,
  (b) $\mathcal{B}_\infty(ab) \subset \mathcal{B}_\infty(b)$;

(ii) there exist $\lambda > 0$, C and $L \in \mathbb{N}$ such that if $|\gamma| \geq L$ and $x, y \in \mathcal{B}_\infty(\gamma)$ then

$$d(\gamma x, \gamma y) \leq Ce^{-|\gamma|\lambda} d(x, y);$$

(iii) there exist constants r_1, r_2 and a function $\tau : \Gamma \to \mathbb{R}$ with $\tau(\gamma) \geq \lambda|\gamma|$ such that for every $\gamma \in \Gamma$ and every $x \in \mathcal{C}_\infty(\gamma)$,

$$B\left(x, r_1 e^{-\tau(\gamma)}\right) \subset \gamma\mathcal{B}_\infty(\gamma) \subset B\left(x, r_2 e^{-\tau(\gamma)}\right).$$

The goal of the subsection is to prove the following result, similar arguments can be found in Guizhen [Cui96] in the context of conjugacies of expanding circle maps.

Proposition 7.2. *Let ρ,$\bar{\rho}$ be as above and assume both admit a Lipschitz compatible cover. If there exists $p \in \partial\Gamma$ such that Ξ has a finite non-vanishing derivative (complex derivative in the Kleinian case) at p then $\Xi|\partial\Gamma$ is bi-Lipschitz.*

We work under the assumptions of Proposition 7.2 and begin its proof with the following lemma. For $\gamma \in \Gamma$ we denote its derivative at $x \in X$ by $\gamma'(x) \in \mathbb{K}$ defined, according our two situations, by:

- $X = \mathbb{S}^1$: the derivative $\tilde{\gamma}'(\tilde{x})$ of a lift of γ to the universal cover $\mathbb{R}$ of $\mathbb{S}^1$, and a lift $\tilde{x} \in \mathbb{R}$ of x, the number $\tilde{\gamma}'(\tilde{x})$ is independent of these choices;
- $X = \partial\mathbb{H}^3$: we fix an arbitrary point $\infty \notin \partial\Gamma$, identify $X - \{\infty\}$ with $\mathbb{C}$ via the stereographic projection and let $\gamma'(x)$ be the standard complex derivative.

Lemma 7.3. *Let $\rho : \Gamma \to \mathrm{Diff}^{1+\nu}(X)$ admit a Lipschitz compatible cover. There exists a constant $\kappa > 0$ and $N \in \mathbb{N}$ such that for all $\gamma \in \Gamma$ with $|\gamma| \geq N$ and $x, y \in \mathcal{B}_\infty(\gamma)$ one has*

$$\big|\log|\gamma'(x)| - \log|\gamma'(y)|\big| \leq \kappa d(x,y)^\nu.$$

Proof. We consider L from Definition 7.1, so that for every $\eta \in \Gamma$ with $|\eta| \geq L$ and $x, y \in \mathcal{B}_\infty(\eta)$ one has

$$d(\eta x, \eta y) \leq Ce^{-|\eta|\lambda} d(x,y). \tag{7.1}$$

Since the action is $\mathrm{C}^{1+\nu}$ we can find a positive K such that for every β with $|\beta| \leq L$ and $u, w \in X$ one has

$$\big|\log|\beta'(u)| - \log|\beta'(w)|\big| \leq Kd(u,w)^\nu. \tag{7.2}$$

We let then $K' = \max\{K, KC^\nu\}$. We begin by showing, by induction on k, that if $|\gamma| = kL$ then for all $x, y \in \mathcal{B}_\infty(\gamma)$, one has

$$\big|\log|\gamma'(x)| - \log|\gamma'(y)|\big| \leq K' \left(\sum_{i=0}^{k-1} e^{-\nu\lambda Li}\right) d(x,y)^\nu. \tag{7.3}$$

Eq. (7.2) gives the base case, so assume that the result holds up to $k-1$. We write $\gamma = \beta\eta$ with $|\beta| = L$, $|\eta| = (k-1)L$. By Definition 7.1 (ib) we have

$$\mathcal{B}_\infty(\gamma) \subseteq \mathcal{B}_\infty(\eta). \tag{7.4}$$

Applying the chain rule gives that for every $u \in X$ one has

$$\log|\gamma'(u)| = \log|(\beta)'(\eta u)| + \log|(\eta)'(u)|$$

and thus, when $x, y \in \mathcal{B}_\infty(\gamma)$,

$$\begin{aligned}
&\big|\log|\gamma'(x)| - \log|\gamma'(y)|\big| \\
&\qquad \leq \big|\log|\beta'(\eta x)| - \log|\beta'(\eta y)|\big| + \big|\log|\eta'(x)| - \log|\eta'(y)|\big| \\
&\qquad \leq Kd(\eta x, \eta y)^\nu + K' \left(\sum_{i=0}^{k-2} e^{-\nu\lambda Li}\right) d(x,y)^\nu \quad \text{(by (7.2) and induction)} \\
&\qquad \leq KC^\nu e^{-|\eta|\nu\lambda} d(x,y)^\nu + K' \left(\sum_{i=0}^{k-2} e^{-\nu\lambda Li}\right) d(x,y)^\nu \quad \text{(by (7.4) and (7.1))}.
\end{aligned}$$

This shows Eq. (7.3) which implies that for $\kappa_0 = K'/(1 - e^{-\nu\lambda L})$, every $\gamma \in \Gamma$ whose word-length is an integer multiple of L, and $x, y \in \mathcal{B}_\infty(\gamma)$ one has

$$\big|\log|\gamma'(x)| - \log|\gamma'(y)|\big| \leq \kappa_0 d(x,y)^\nu.$$

To conclude the lemma we consider an arbitrary γ with $|\gamma| = mL + t$ and $t < L$. We write $\gamma = \beta\eta$ with $|\beta| = mL$. By Definition 7.1 (ia) it holds

$$\eta\mathcal{B}_\infty(\gamma) \subset \mathcal{B}_\infty(\beta). \tag{7.5}$$

Applying the chain rule gives then

$$\begin{aligned}\left|\log|\gamma'(x)| - \log|\gamma'(y)|\right| &\leq \left|\log|\beta'(\eta x)| - \log|\beta'(\eta y)|\right| + \left|\log|\eta'(x)| - \log|\eta'(y)|\right| \\ &\leq \kappa_0 d(\eta x, \eta y)^\nu + K d(x,y)^\nu \quad \text{(by (7.2) and (7.5))} \\ &\leq \left(\kappa_0 C^\nu e^{-mL\lambda} + K\right) d(x,y)^\nu \ \text{(by (7.1))}\end{aligned}$$

so taking $\kappa = K + \kappa_0 C^\nu e^{-L\lambda}$ we conclude the proof of Lemma 7.3. □

Proof of Proposition 7.2. Let $p \in \partial\Gamma$ be such that Ξ has a finite non-vanishing derivative. Fix a geodesic ray $(\alpha_n)_0^\infty$ through the identity with $\alpha_n \to p$. By definition for all n one has $p \in \alpha_n \mathcal{C}_\infty(\alpha_n)$. Without loss of generality we may also assume that

$$p = 0 = \Xi(0)$$

and we may write the derivative as the incremental limit

$$\Xi'(0) = \lim_{y\to 0} \frac{\Xi(y)}{y} \in \mathbb{K} - \{0\}.$$

For each n we let $s_n = r_1 e^{-\tau(\alpha_n)}$, so that by Definition 7.1 (iii),

$$B(0, s_n) \subset \alpha_n \mathcal{B}_\infty(\alpha_n).$$

We consider the scaling map

$$g_n : B(0,1) \to \alpha_n \mathcal{B}_\infty(\alpha_n)$$

defined by $g_n(z) = s_n z$.

Let a_n be an arbitrary point at distance s_n from 0 and let $\tilde{s}_n = \Xi(a_n)$. Observe that since Ξ is differentiable at zero, for n big enough the image $\Xi(B(0,s_n))$ is coarsely a ball around zero of size comparable to that of $\overline{\alpha}_n \mathcal{C}_\infty(\overline{\alpha}_n)$, and in particular we can assume, since the cover $\{\mathcal{B}_\infty(\overline{\gamma})\}$ is Lipschitz compatible (Definition 7.1 (iii), that $\Xi(B(0,s_n))$ is contained in $\overline{\alpha}_n \mathcal{B}_\infty(\overline{\alpha}_n)$. Furthermore we deduce that there exist positive constants d, D such that for every n

$$d < \frac{\overline{r}_2 e^{-\bar{\tau}(\overline{\alpha}_n)}}{|\tilde{s}_n|} < D.$$

Here we denote by $\overline{r}_i, \overline{\lambda}, \bar{C}, \bar{\tau}$ the constants and function associated to the Lipschitz compatible cover $\{\mathcal{B}_\infty(\overline{\gamma})\}$ for the action $\overline{\rho}$. We consider the scaling map

$$\tilde{g}_n : B(0,D) \to B(0, |\tilde{s}_n| D)$$

by $z \mapsto z\tilde{s}_n$.

Since $s_n \to 0$ and $\Xi'(0) \notin \{0, \infty\}$ exists, the composition

$$\tilde{g}_n^{-1} \Xi g_n(z) = \frac{\Xi(z s_n)}{\tilde{s}_n} \cdot \frac{s_n z}{s_n z} = \frac{\Xi(z s_n)}{s_n z} \cdot \frac{s_n}{\tilde{s}_n} \cdot z = \frac{\Xi(z s_n)}{s_n z} \cdot \frac{s_n}{\Xi(s_n)} \cdot z$$

converges uniformly on compact subsets to the identity map.

On the other hand, one has

$$\tilde{g}_n^{-1} \Xi g_n = \tilde{g}_n^{-1} \overline{\alpha}_n \Xi \alpha_n^{-1} g_n.$$

We now study the maps $f_n := \alpha_n^{-1} \circ g_n$ and $\tilde{f}_n := \tilde{g}_n^{-1} \circ \overline{\alpha}_n$. Since the coverings $\mathcal{B}$ and $\overline{\mathcal{B}}$ are finite, we can assume, up to extracting a subsequence that there exists sets $\mathcal{B}_\infty \in \mathcal{B}$, $\overline{\mathcal{B}}_\infty \in \overline{\mathcal{B}}$ so that, for every n, $\mathcal{B}_\infty(\alpha_n) = \mathcal{B}_\infty$ (resp. $\mathcal{B}_\infty(\overline{\alpha}_n) = \overline{\mathcal{B}}_\infty$).

Observe that for every $x \in B(0,1)$ one has

$$\log|f_n'(x)| = \log\left|\left(\alpha_n^{-1}\right)'(g_n x)\right| + \log|s_n| = -\log\left|\alpha_n'\left(\alpha_n^{-1} g_n x\right)\right| + \log|s_n|.$$

Now by definition of g_n, we have that $g_n x \in \alpha_n \mathcal{B}_\infty(\alpha_n)$ and thus $\alpha_n^{-1}(g_n x) \in \mathcal{B}_\infty(\alpha_n)$. For n large enough we can apply Lemma 7.3 to α_n to obtain κ so that for every pair $x, y \in B(0,1)$ it holds

$$\left|\log|f_n'(x)| - \log|f_n'(y)|\right| \leq \kappa d(x,y)^\nu.$$

We conclude that the family of maps (f_n) is uniformly bi-Lipschitz on $B(0,1)$ and thus, since $(f_n 0)$ is bounded, Arzela–Ascoli's theorem applies to give a subsequence (still denoted by f_n) that converges to a bi-Lipschitz map f defined on $B(0,1)$.

A similar reasoning applies to the maps $\tilde{f}$ defined on $\overline{\mathcal{B}}_\infty$, and we obtain that, about 0, Ξ can be written as a composition of bi-Lipschitz maps and is thus bi-Lipschitz. Let U be the open neighborhood where Ξ is bi-Lipschitz, then for every $\gamma \in \Gamma$, Ξ is bi-Lipschitz on γU with possibly different constants. However, by compactness we can cover $\partial\Gamma$ by finitely many open sets such that Ξ is bi-Lipschitz restricted to each one of them. A priori the bi-Lipschitz constants of Ξ depend on each open set but we may choose them to work simultaneously for these finitely many translates of U. It remains then to check the bi-Lipschitz property on points which are uniformly far apart, i.e. whose distance is $\geq \varepsilon$ for some $\varepsilon > 0$. However, the function

$$(x,y) \mapsto \frac{d(f(x),f(y))}{d(x,y)}$$

is continuous and non-vanishing on $\{(x,y) : d(x,y) \geq \varepsilon\}$. Since the latter space is compact we obtain the bi-Lipschitz property for points which are far apart, as desired. □

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